

Muhammad Asif Ali Khan

Federated Learning Researcher · AI Engineer · MS Data Science (Silver Medalist)

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MS Data Science graduate (Silver Medalist) with a foundation in pure mathematics, specializing in federated learning. My research quantifies client contribution with Shapley-value credit assignment and designs aggregation that stays fair and robust under non-IID data, with two papers under review at Springer *Machine Learning*. Experienced in teaching machine learning and mathematics. Seeking a PhD position in trustworthy federated learning or a lecturer role in AI and data science.

RESEARCH INTERESTS

Federated Learning, AI, Computer Vision, Non-IID Learning, Privacy-Preserving Machine Learning, Trustworthy AI

EDUCATION

MS Data Science *Shifa Tameer-e-Millat University, Islamabad*

May 2024 – Mar 2026

- CGPA 3.64/4.0 · **Silver Medalist**
- Research: *A Contribution-Aware Aggregation in Federated Learning*
- Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistical and Mathematical Methods for Data Science, Data Distribution Processes, Tools and Techniques

BS Mathematics *International Islamic University Islamabad*

Sep 2019 – Oct 2023

- CGPA 3.05/4.0
- Coursework: Linear Algebra, Calculus, Topology, Functional Analysis, Complex Analysis, Differential Geometry, Ring Theory, Module Theory

PUBLICATIONS

- “SimFL-FWShapley: Efficient and Drift-Aware Shapley-Based Credit Assignment for Federated GBDT Learning.” *Machine Learning*, Springer. (**Under review**)
- “FedCAWS: An Efficient Amortized Learning-Based Contribution-Aware Aggregation Framework for Federated Learning.” *Machine Learning*, Springer. (**Under review**)

RESEARCH EXPERIENCE

MS Research: A Contribution-Aware Aggregation in Federated Learning

Shifa Tameer-e-Millat University, Islamabad

- Designed a contribution-aware federated learning framework that improves model stability under non-IID data and partial client participation.
- Combined FedProx-style proximal local training with FW-Shapley contribution estimation and Beta-Shapley coalition sampling for scalable client utility measurement.
- Built dynamic aggregation weights that amplify high-utility client updates and suppress low-utility or harmful ones.
- Achieved smoother convergence and better generalization than FedAvg, FedProx, SFedAvg and GreedyFed on MNIST and Fashion-MNIST.

TEACHING EXPERIENCE

Machine Learning Instructor *Online teaching*

4 months

- Taught machine learning from data preparation and model building to evaluation, with hands-on Python and Scikit-learn sessions.

Private Tutor, Intermediate (FSc) *Home tuition*

1.5 years

- Taught FSc subjects one-on-one, adapting explanations to each student's level and preparing students for board examinations.

Courses I can teach: Machine Learning, Deep Learning, Natural Language Processing, Statistics, Linear Algebra, Data Science.

PROFESSIONAL EXPERIENCE

- AI Engineer Intern Eziline Software House, Islamabad Feb 2026 – Jul 2026
- Built and deployed a Flask web app that classifies abusive and non-abusive Roman Urdu comments in real time using TF-IDF features and multiple classifiers (88% accuracy).
 - Designed a 2D U-Net for brain tumor segmentation on multi-modal MRI (Dice 0.85), with IoU evaluation, prediction overlays and a Streamlit interface.
 - Developed a Streamlit NLP app for real-time emotion detection in English text (90% accuracy) with confidence scores.
 - Built an AI voice assistant using GroqCloud LLM, Whisper speech-to-text and gTTS for browser-based voice interaction.
 - Fine-tuned YOLOv5s on PASCAL VOC (20 classes) for real-time object detection, with webcam and video inference and ONNX/TorchScript export for edge deployment.

SELECTED PROJECTS

- Bank Churn Prediction** (Oct 2025): Deep neural network predicting customer churn with 88% accuracy, explained with SHAP.
- Student Stress Level Classifier** (Jan – Mar 2026): Three-class model with 90% accuracy, deployed as a Streamlit dashboard.
- Book Recommendation System** (Aug 2024): Hybrid collaborative and content-based recommender with NLP features, deployed with Flask and Streamlit.
- House Price Prediction** (Jun 2025): Regression with feature engineering and tuned Gradient Boosting and Random Forest models.

CERTIFICATIONS

- Develop Generative AI Applications: Get Started**, IBM / Coursera (Aug 2026)
- The Future of AI: Navigating AGI and ASI**, Special Competitive Studies Project / Coursera (Aug 2026)
- Python for Data Science, AI & Development**, IBM / Coursera (Jul 2026)

TECHNICAL SKILLS

Languages	Python, R
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn; CNN, RNN, LSTM, Transformers, U-Net
NLP & LLMs	BERT, GPT, LangChain, NLTK, Word2Vec, GloVe, TF-IDF
Computer Vision	YOLOv5, image segmentation, ONNX / TorchScript export
Data & Visualization	Pandas, NumPy, PyCaret, Matplotlib, Seaborn, Plotly, ggplot2
Tools & Deployment	Flask, Streamlit, Jupyter, Google Colab, VS Code

HONORS

- Silver Medal**, MS Data Science, Shifa Tameer-e-Millat University (2026)

REFERENCES

Available upon request.